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Abstract

The phase of a signal is a non-trivial quantity. It is therefore often ignored in favor of signal magnitude. However, phase conveys more information regarding signal structure than magnitude does, especially in the case of images. It is therefore imperative to use phase information in various signal/image processing schemes, as well as in computer vision. This is true for global phase and, even more so, for local phase. The latter is sufficient for signal/image representation, while totally ignoring the magnitude information. The implementation of localized methods requires substantial computation resources. Thanks to the major progress in available computing resources during the last decade, the implementation of localized methods has become feasible. Thus, there is a growing interest in localized approaches, including those that incorporate phase in, both theory and application. We address the importance of phase, in image processing with special emphasis on its application in edge detection and segmentation.

Introduction

One of the most important and widely used tools for image representation and analysis is the spatial frequency transform, which can be represented in terms of magnitude and phase. The importance of phase in images was first shown in the context of global phase [7]. Since usually the content of a signal is not stationary, the localized frequency analysis has become an important and powerful tool in signal representation [9, 10]. In order to deal with such non-stationary signal, it is advantageous to analyze the signal frequency and spatial information simultaneously, with maximal possible resolution in both position and frequency. The joint resolution is however, limited by the uncertainty principle. The drawbacks of the Fourier transform (such as lack of spatial localization), and the tools to overcome these limitations (spatial-frequency analysis schemes) were discussed in [11].

The importance of phase information in images has inspired its implementation in various tasks such as edge and corner detection [12], image segmentation [13, 14], and more... Phase is highly immuned to noise and contrast distortions- features desirable in image processing.

The report is organized as follows. We begin with discussion of the importance of phase demonstrating it by means of several examples. We then proceed with two primal applications in image processing tasks of segmentation and edge detection, where novel phase-based methods show good performance compared to classical methods. Subsequently we introduce our own edge detector- the Local Phase Quantization error (LPQe), and address its performance and possible applications [15]. Finally we describe the Rotated Local Phase Quantization (RLPQ) which achieves controlled image primitives deprival (presented in this report for the first time). Several RLPQ applications are also discussed.

Global and Local Phase

Global Phase

We first wish to examine qualitatively- which of the two, magnitude or phase, carries more visual information. This can be most vividly demonstrated by the following experiment: the Fourier components- (phase and magnitude) are generated for the two images of same dimensions, and then swapped (see figure 1), whereby the reconstructed images appears to be more similar to the one whose Fourier phase was used in the reconstruction. This experiment was previously suggested by Oppenheim in [7] and elsewhere.

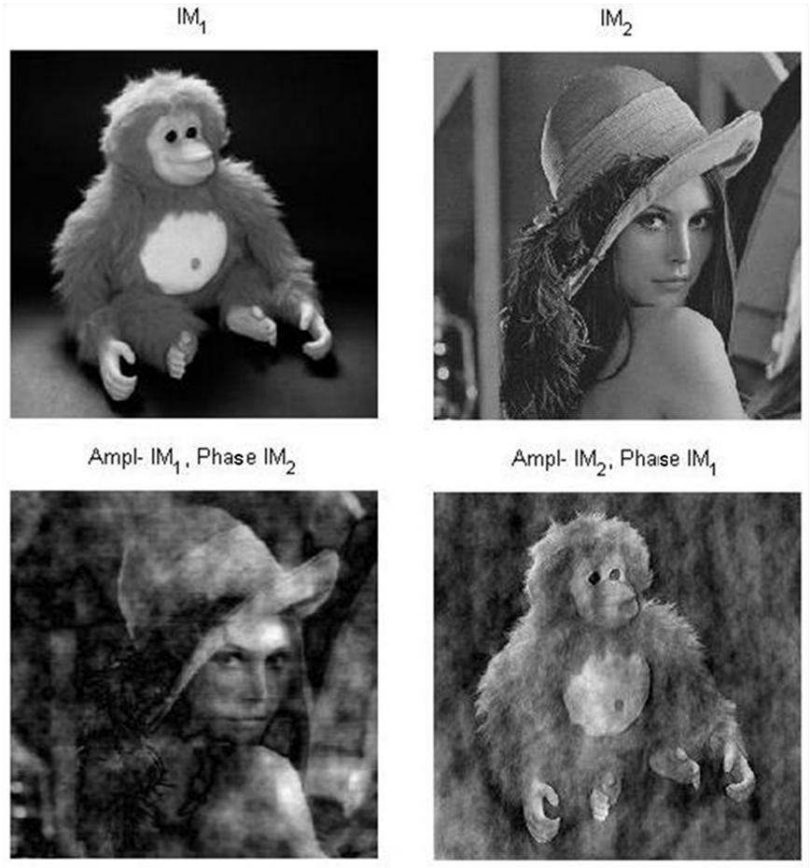


Figure 1: Swapping the Fourier phase and magnitude in images. Top left - original Lena image. Top right- original monkey image. Bottom left- IFT of Lena phase and monkey magnitude. Bottom right- IFT of monkey phase and Lena magnitude.

We have repeated the same experiment for a 1D signal- voice in this case: two different sentences, pronounced by individuals of different gender were recorded. The signals' phase and magnitude were swapped. The resulting sentences were played to human listeners- which were able to understand the meaning of the sentence, as well as to identify the gender of the speaker. Thus, it appears that most of the signal's information is carried by its phase in 1D case as well. The effect of using a swapped magnitude resulted in appearance of noise, in a manner similar to the 2D case.

The reader is encouraged to review figure 2 and to examine the spectrograms similarity, or to use this link for the audio files (click images to download the file) in order to evaluate the importance of phase in human voice signals.

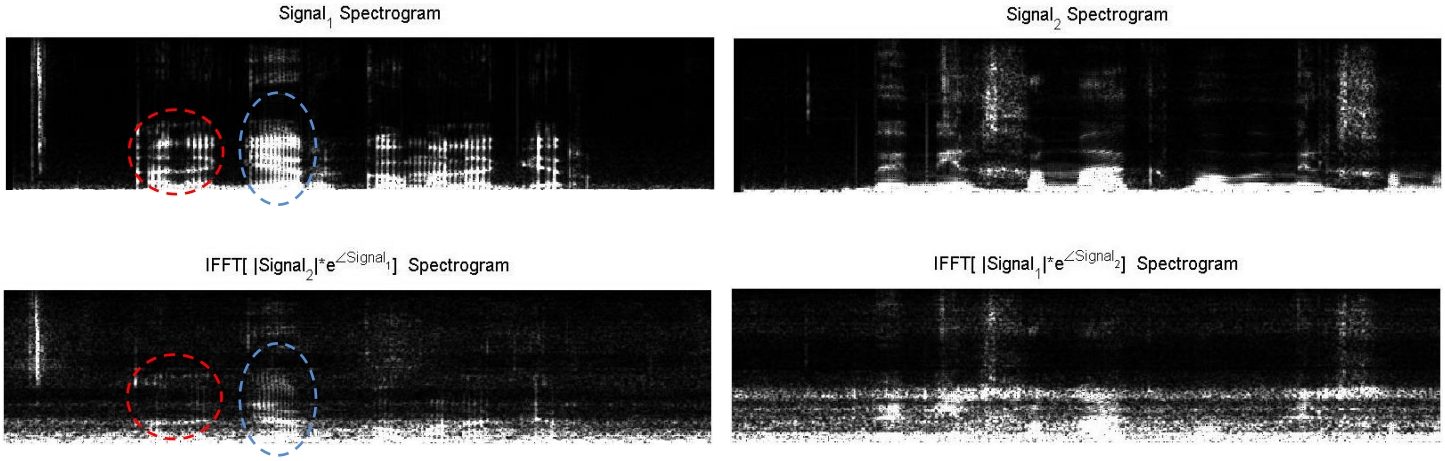


Figure 2: Exchanging the Fourier phase and magnitude in voice. Top left - woman voice spectrogram. Top right- man voice spectrogram. Bottom left- spectrogram of woman voice phase and man voice magnitude. Bottom right- spectrogram of man voice phase and woman voice magnitude. Both reconstructions are primarily dominated by Fourier phase, and not the magnitude.

Next, let us compare the global phase and magnitude by reviewing their distribution in a realistic image (Lena image in this case).

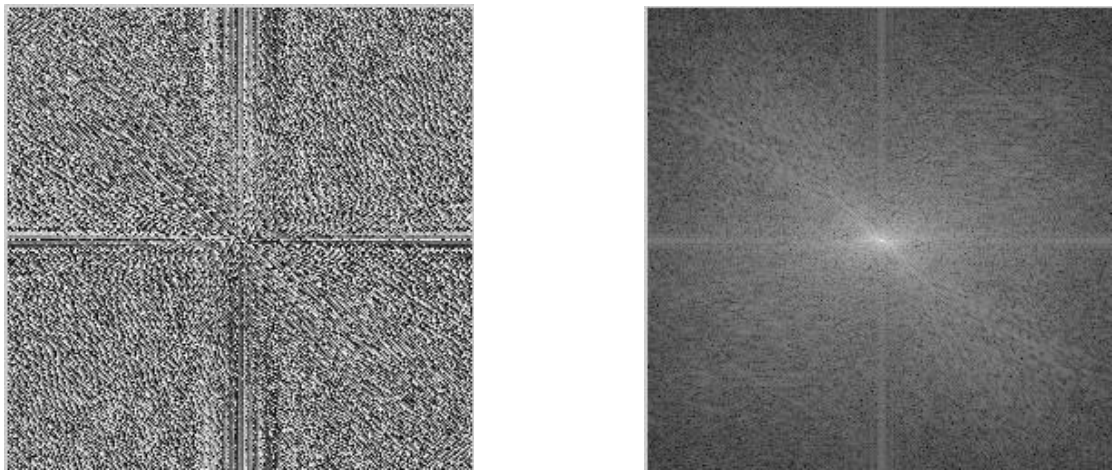


Figure 3: Global Fourier phase and magnitude of Lena image. Left- global Fourier phase. Right- global Fourier magnitude. It can be seen that the magnitude decays, and most of it energy is concentrated in the middle, while the phase is distributed through all frequencies.

A quick glance reveals that while phase is almost equally distributed across the spectrum and the entire range of frequencies is exploited, the magnitude decays in an exponential manner with increasing frequency. A 3D plot is depicted in figure 4, for a more detailed view.

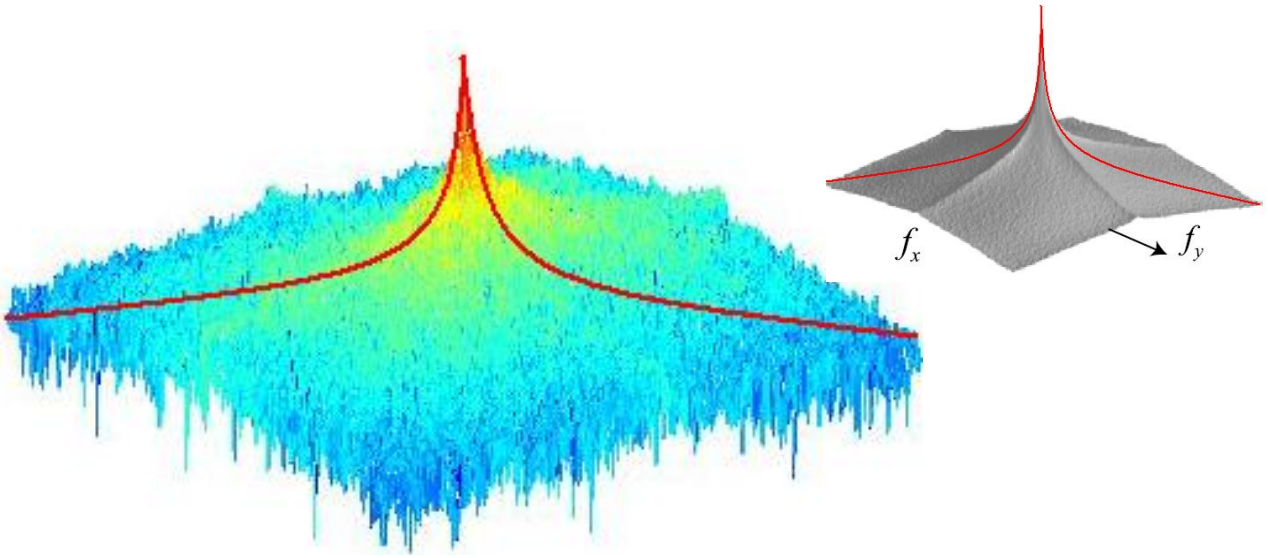


Figure 4: Lena image spectrum and natural images statistical average of spectra. Left- global "Lena" Fourier magnitude. Right- global Fourier magnitude achieved from average of various natural images (adopted from [1]). Spectra similarity is clearly seen.

In fact, the spectral distribution presented in figure 4 is common to all natural images [1]. This strengthens our observation that the information that differentiates between images is not encoded in the Fourier spectra, but is encoded in the phase.

Local Phase

After describing the importance of global phase, as compared to the global magnitude, we turn to local phase. Global Fourier analysis provides information on the frequency contents of the whole signal. Assuming the signal is non stationary, these contents will vary in time/location, and thus the global Fourier transform analysis is ill sufficient. As we are sometimes interested in the frequency contents in a certain part of the signal we must use localized schemes. Unfortunately, one cannot determine both spatial position and frequency with infinite accuracy. According to the Time-Frequency Uncertainty Principle (derived from the Heisenberg uncertainty principle) time and frequency accuracy product is limited, as described in the following equation:

$$\Delta_t \Delta_f \geq 1/2 \qquad (1)$$

The analysis of the combined frequency-spatial space can be achieved using various tools, such as Short Time Fourier Transform (STFT), Gabor Transform (GT) and Wavelets Transform. One of the differences between the above schemes is the Time-Frequency Uncertainty each of them capable to achieve.

As with the global case, we wish to demonstrate that localized phase can be used for signal analysis, and that in some cases it outperforms local magnitude based methods. Moreover, it has been shown that the

localized (Gabor)-phase is sufficient for image reconstruction and that the row data of Gabor phase depicts the contour information starting from the first iteration of the reconstruction process [4]. This fact alone implies that localized phase is substantial for signal analysis, as it carries all necessary signals information.

We will compare the difference signal reconstruction schemes, based on partial Fourier Transform information, both local and global. A wide range of papers [2-4, 16, 17] are devoted to signal reconstruction using partial spatial and frequency data divided to Fourier phase or magnitude. These can be handy in cases where not all FT information is available (like with SAR images and X-ray crystallography) or when it is degraded. We wish to demonstrate that the use of local features allows better algorithm performance: faster convergence, or usage of less a priory known data. We also intend to demonstrate that phase based algorithms sometimes result in a superior outcome compared to magnitude based ones.

We will address iterative schemes, as the closed form solutions demand solving a large set of linear equations, which in turn involves inversion of appropriate matrices. Those matrices inversion is impractical for images of dimensions above 16X16 pixels.

A Global Magnitude reconstruction scheme presented in [2] can be seen in the following figure 5. The proposed methods allow the reconstruction of a signal using at least 25% of the image (half of the signal in each dimension) and its Fourier magnitude. As the reader can see, the reconstruction is achieved by an iterative detection of the unknown part of the signal. The authors of [2] report a decent signal reconstruction after 50 iterations. In their next paper [3] the authors propose a Local magnitude-based image reconstruction method. While the reconstruction process converges faster (fever stages for same quality- see figure 7), it demands more computations. Out of those stages, the last one, for example, will demand the same number of iterations as the whole Global Magnitude based method. On the other hand the number of spatial points to be known in advance drops to 1 as opposed to the ~25% needed by the Global Magnitude based method. The proposed method consists of application of the Global Magnitude based method to an increasing part of the original signal, until whole signal reconstruction is achieved. A graphical description can be seen in figure 7.

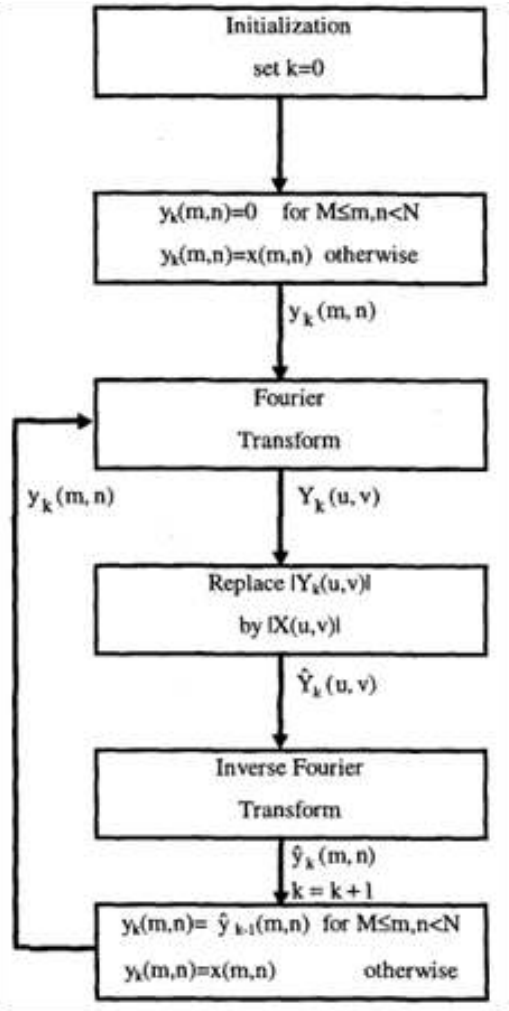


Figure 5: A flowchart of image reconstruction from global magnitude. Diagram adopted from [2].

As can be seen from the following figure, the scheme is applied to a sub signal of the dimensions of [2^k, 2^k], where k is the iteration number (X_k is the appropriate label on the figure). An N by M image will demand $\lceil \log_2(\max[M,N]) \rceil$ applications of the Global magnitude scheme (which is iterative too) to a sub image of [2^k, 2^k] dimensions.

It can be seen that in a case where no spatial data except *x(0,0)* is available, only the localized magnitude-based scheme will be sufficient for image reconstruction, though a computational price is to be paid.

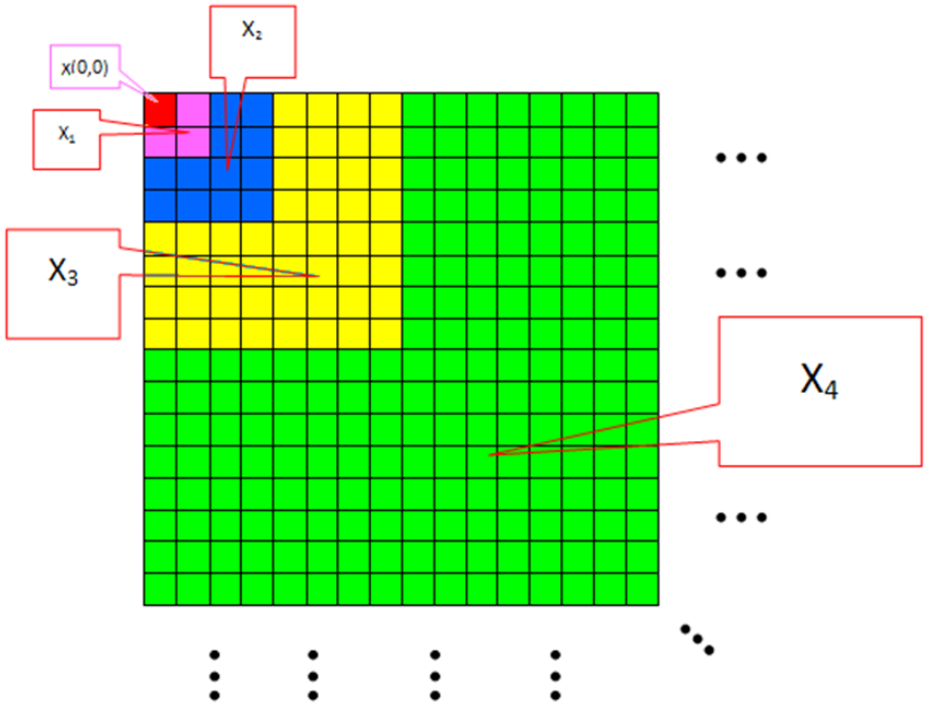


Figure 6: Description of a Local magnitude-based image reconstruction iterative algorithm from [3].

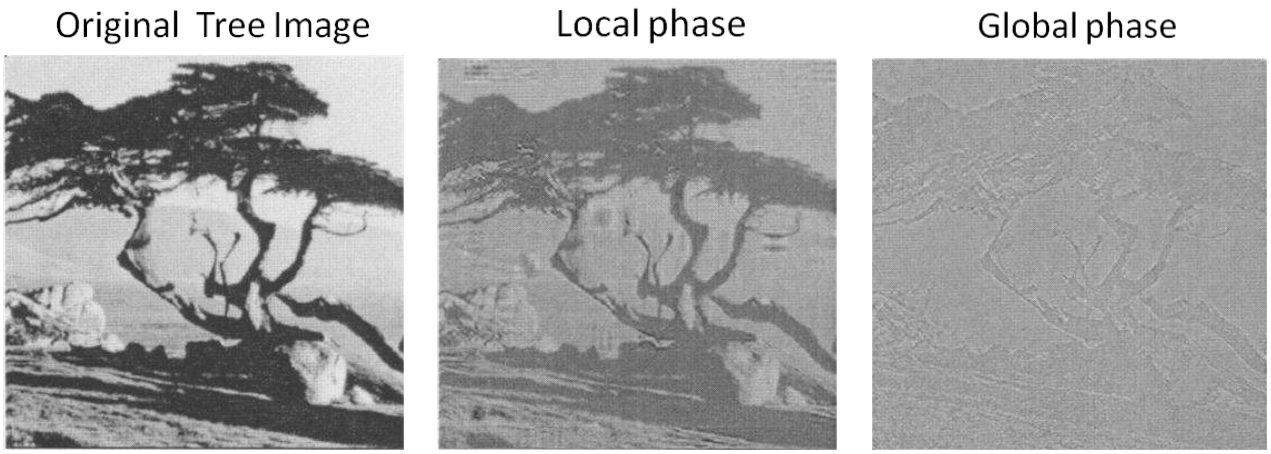


Figure 7: A comparison of images reconstructed from local and global phase (adopted from [4]). Left - original tree image. Middle- single iteration of local phase-based algorithm reconstructed image. Right- single iteration of global phase-based algorithm reconstructed image.

Local phase applications

We are motivated to use the local phase due to some of it's good qualities- immunity to illumination change, immunity to zero phase types of noise and the fact its value is limited [-π: π]. Therefore we assume and plan to demonstrate that use of local phase based schemes results in better results.

Segmentation

Segmentation is one of the most fundamental and basic image analysis tasks. Its main goal is to divide the image into meaningful parts (segments) - based on common characteristics. Image segmentation is usually used for object extraction (i.e., separation from the background) and its boundary detection. Some of the practical applications of image segmentation are: objects detection in satellite images [18-20], person identification (using face, fingerprint or iris [21, 22] recognition), medical imaging (detection of tumors and other pathologies, measuring tissues dimensions, computer-guided surgery, etc…) and even traffic control systems.

Various techniques and algorithms were proposed for image segmentation (see [23] for comparison of various methods). Segmentation usually relies on either edge detection or region identification. A common image segmentation method is based on unsupervised segmentation via clustering. It utilizes the magnitude or real part of the output of the Gabor Filter Bank [24-27]. The segmentation scheme described in [13] is based on both the Gabor magnitude and phase. The authors claim that each texture has a unique distribution of both Gabor magnitude and phase responses. The latter is calculated from the difference between adjacent pixels in *x*, and *y* directions. The appropriate statistical values $[\sigma_x, \sigma_y, \mu_x, \mu_y]$ are calculated for each texture from a predefined textures bank.

During the analysis process, each of the tested textures parameters is evaluated, and used to determine the most similar texture from the predefined textures bank.

The feature space we adopt from [13] is based on statistical properties of Gabor phase distribution. We propose the following scheme: each element of the feature matrix will be derived from the statistical distribution of its neighbors. Thus, assuming that the distribution of the Gabor features of each texture is unique (a signature), the statistical parameters of the feature matrix will be sufficient for a successful segmentation.

The elements of the phase-based feature vector of pixel *(x,y)* are: $[\sigma_x(x,y), \sigma_y(x,y), \mu_x(x,y), \mu_y(x,y)]$. Here $\sigma_x(x,y)$ is the standard deviation of the phase difference in *x* direction in a N-by-N neighborhood of the filtering results, around pixel *(x,y)*. Likewise $\mu_y(x,y)$ is the mean value of the phase difference in *y* direction in a N-by-N neighborhood of filtering results, around pixel *(x,y)*. The rest is clear.

At following figure 8 we present some textures from the Brodatz textures album [28], and their distribution parameters vectors $[\sigma_x, \sigma_y, \mu_x, \mu_y]$, in order to check whether the distribution of the responses of the Gabor phase is unique for those textures. A Gabor wavelet filter bank of moderate dimensions (two scales and four orientations) was applied in our analysis.

A short glance at figure 8 reveals the differences in the statistical values of some textures. Assuming these differences will be sufficient for a successful segmentation, we perform image segmentation using a phase-based feature space composed of statistical parameters vector $[\sigma_x, \sigma_y, \mu_x, \mu_y]$ of Gabor phase response difference.

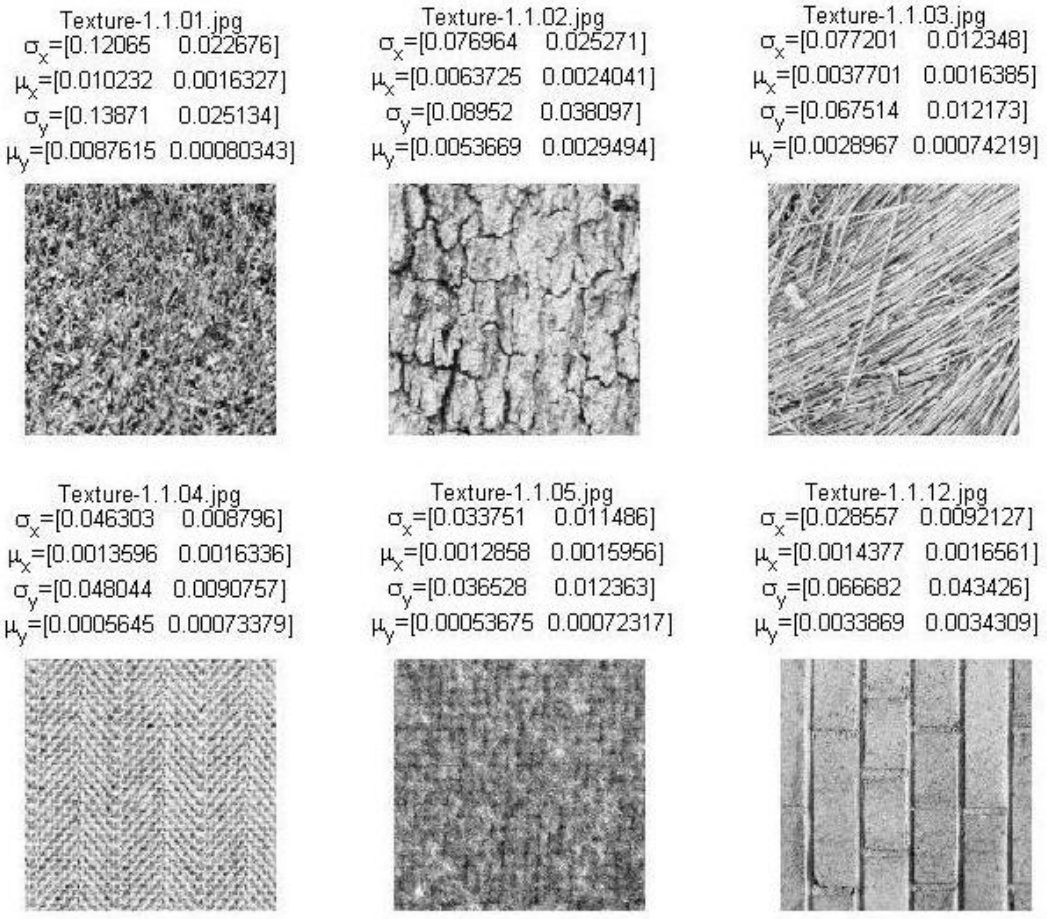


Figure 8: Phase statistical data of Brodatz textures with their Gabor phase mean and variance values

The segmentation results can be seen in the following table (best grades are marked with red):

Feature space \ Ex. No'	1	2	3	4	5	6	7	8	9	*Average Method Grade*
Magnitude only	71.51	96.50	67.62	49.59	69.73	61.91	71.18	43.54	60.38	65.77
Phase only	94.33	82.71	63.80	93.79	79.71	71.44	58.92	58.76	58.85	73.59
Combined (Phase & Mag.)	71.61	95.84	81.54	95.23	77.77	75.78	76.95	37.90	69.35	75.77
Average Mosaic Grade	79.15	91.68	70.99	79.54	75.74	69.71	69.02	46.73	62.86	71.71

The above table results are ambiguous. Nevertheless the quality of the segmentation achieved via use of local phase information is better in most cases that one achieved based on magnitude solely. The authors of

[13] proposed a weighted feature space, resulting from multiplication of the Gabor magnitude with the Gabor phase, which resulted in a better segmentation in most cases. Our goal here, however was merely to demonstrate that phase solely is sufficient for proper segmentation, thus we are mainly interested in pure phase-based segmentation. Segmented images and additional data can be seen at the web using the following link.

Edge detection

Edge detection plays a central role in image analysis. For example, the edge map of a scene is sufficient for its understanding by a human observer. Technical instructions are often illustrated by line drawings, for better clarity. While many familiar edge detection schemes are gradient based, they seem to be unable to effectively deal with noise or changes in contrast.

In this section we introduce phase-based schemes for edge detection, and compare them to the well known Canny operator [29]. The phase-based schemes rely on the observation that at edges the phase has a special, ordered behavior that is significantly different from its behavior at non edge locations.

Phase Congruency (PC)

The following phase-based edge detection method was introduced by Peter Kovesy [12] in 1999. It is adopted by more and more researches [29-40] for various applications. Phase Congruency is closely related with the "Local Energy" concept, coined by Morrone and Owens in [41]. The general idea is that various signal features are present at locations where Fourier components are maximally in-phase.

According to its definition in [41], PC is the local minima of the spread of the phase of the various Fourier components of a signal. In other words, in order to calculate PC, we have to calculate the spread of the phases of all the frequency components of the Fourier transform of the signal, $FT\{x(t)\}$ for every point in the signal.

Let us now try to detect points of minimal phase spread, according to the following equation:

$$PC = \frac{\left|\sum X(w)\right|}{\sum\left|X(w)\right|}. \qquad (2)$$

The numerator in the equation is the magnitude of the vector sum. The denominator is the sum of magnitudes of the frequency components. It is obvious that according to the above eq. (2) $0 \le PC \le 1$. PC equals 1 if and only if all the phase components are equal, that is- the phase spread is zero. In figure 9 we compare several edge detector schemes (phase- and gradient-based) for different types of 1D edges.

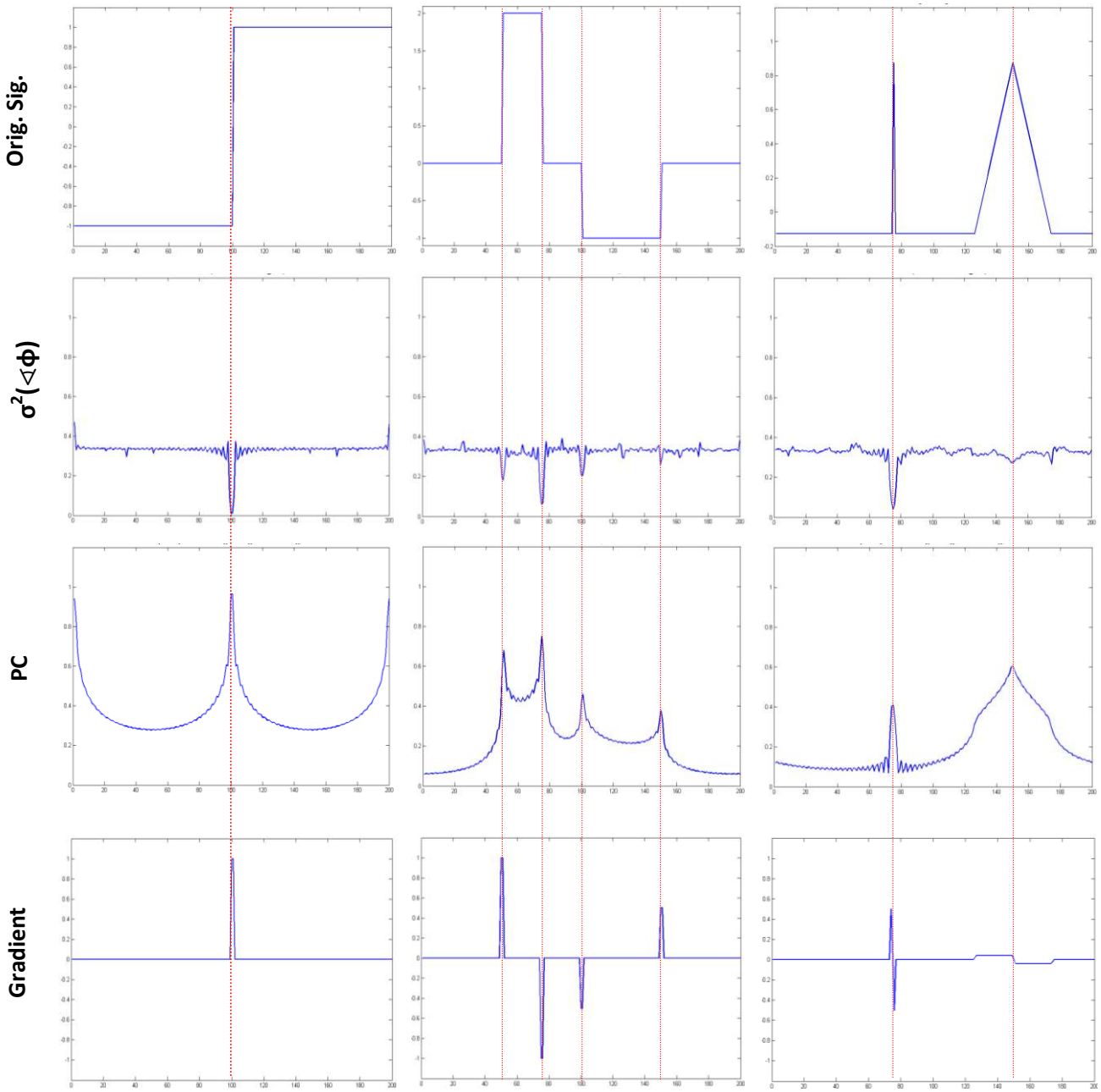


Figure 9: 1D edges detection. Top row- original signals: leftmost- single step, middle - positive and negative windows, leftmost- roof and edge. Columns (top-down order): phase variance, Phase Congruency, Analytic Signal and gradient. The dashed red lines mark the edges locations across different graphs for better clarity.

In his paper [12] the author proposed a method to evaluate the phase congruency of two dimensional signals. The edge detector based on this scheme has several advantages: it detects various features in a desired direction, regardless of the image illumination, it is capable to deal with noise, it is a dimensionless quantity varying from 0 to 1 and finally it was reported to be consistent with the human visual system. Due to its good qualities, the method described in [12], was adopted by researchers for various applications [29-40]. We shall briefly describe this important edge detection scheme. The author proposes to use a complex Gabor wavelets filter bank. The "real" and "imaginary" components of the filters are used for "Local Energy" calculation by summing up the filtering results for each scale. The 1D PC scheme is extended to two

dimensions, as the Gabor filter bank is composed of filters with varying orientations and scales (see figure 10). Each orientation is treated as a 1D case, and the 2D-PC is calculated by summing up the 1D-PC's. In addition, the 2D-PC is enhanced in order to deal with noise and to incorporate a higher frequency spread, but these are of less interest in our research. The following equation describes how 2D-PC can be evaluated:

$$PC(x) = \frac{\sum_{O} \left(E_O(x) - T_O \right)}{\varepsilon + \sum_{O} \sum_{S} A_{O,S}(x)} \quad . \tag{3}$$

Here *O* and *S* stand for Gabor filter orientation and scale respectively, ε prevents division by zero and T_O is the total noise influence for the orientation *O,* applied for noise compensation. The edge maps achieved using 2D PC are presented at figure 13.

While the results achieved using the above scheme are impressive, we believe that the analysis of a 2D signal by its 1D projections is not optimal. One way to deal with the issue is by using the radial Hilbert transform [42, 43]. In the next paragraph we present a somewhat different and interesting approach.

The following figure 10 demonstrates the definition of the 2D PC through several 1D projections.

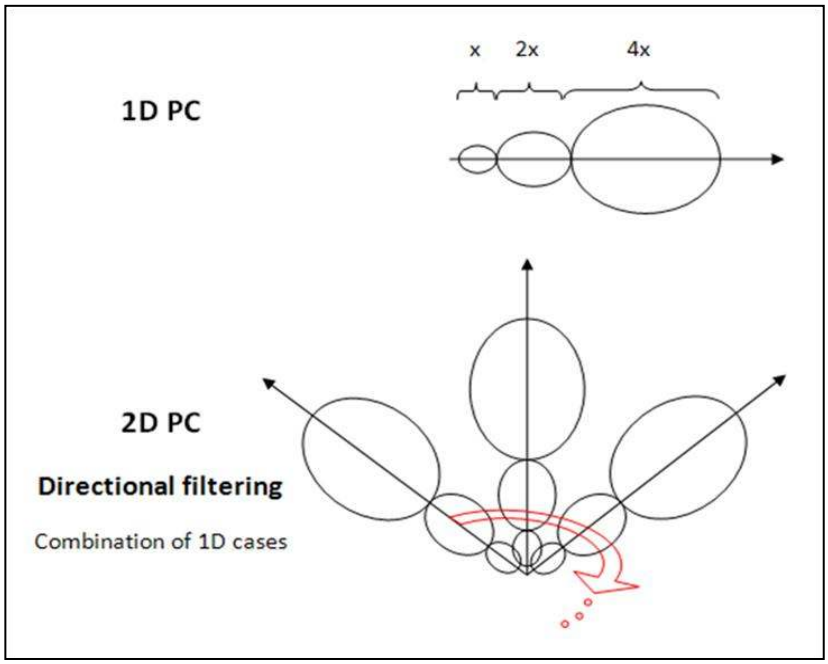


Figure 10: Top- filters needed for 1D PC calculation. Bottom- filters needed for 2D PC calculation. The ellipses on the graphs describe the scale space filters used.

We wish to demonstrate the advantages of interpolating PC in an existing scheme. The algorithm we wish to improve is the classical intensity based active contours snakes, that was first introduced in [44]. The "snakes" algorithm is used to detect objects in an image. This is an iterative algorithm that converges when the "snake" lies on the object's borders. The algorithm described in [44] is gradient-based. Though usually the "snakes" algorithm works fine, under certain conditions it fails. One such condition is an environment of varying illumination. One of the good qualities of the PC we have mentioned above, is its ability to deal with changes in the intensity of the environment. We have implemented the classical snakes algorithm and a PC

based one and applied it to a problematic image that exhibits an illumination varying background (figure 11). The following figure demonstrates that replacing the gradient with PC allows the enhanced "snakes" algorithm to succeed where it originally failed. A video can be seen on the web using this link.

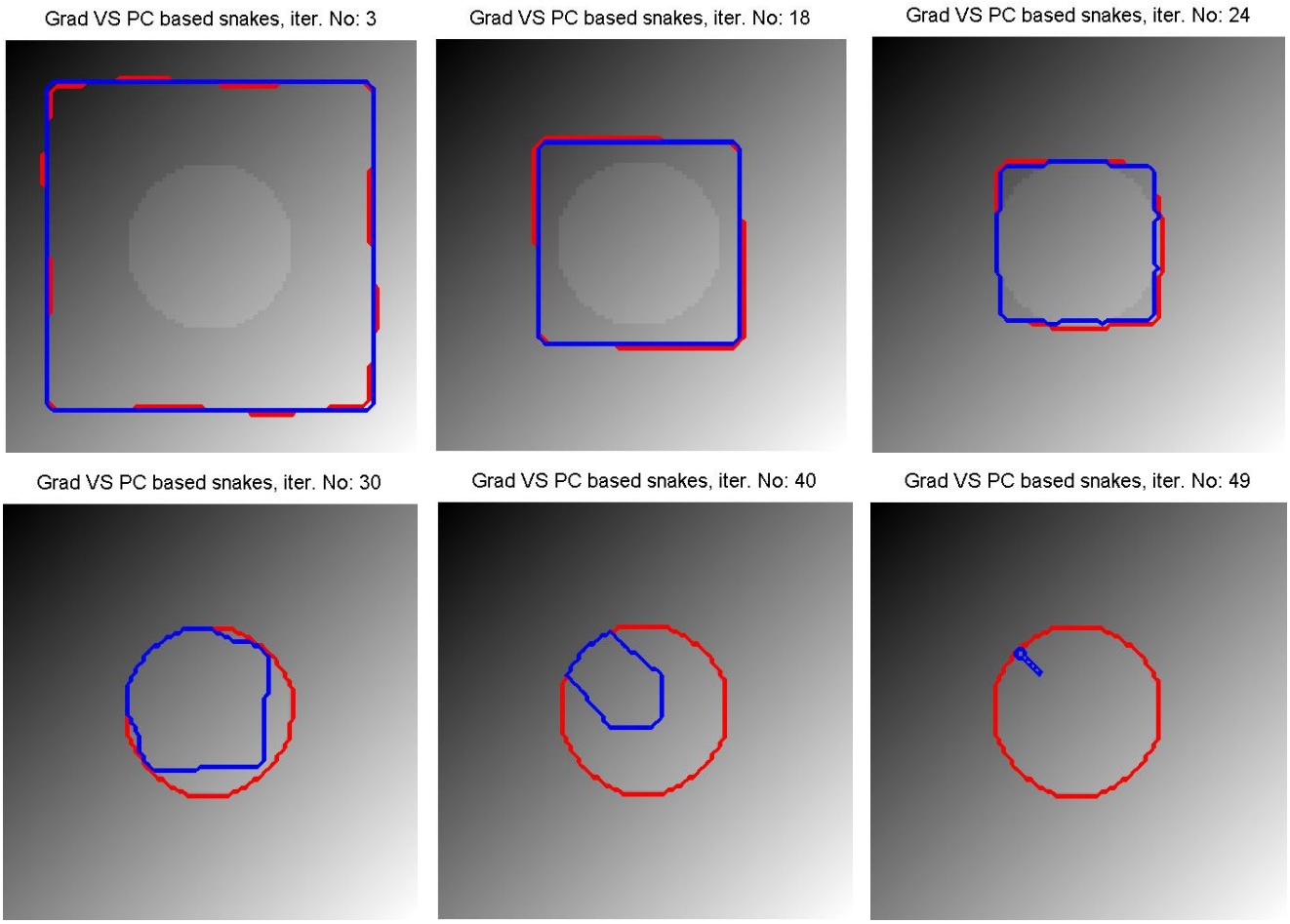


Figure 11: Gradient versus PC-based "snakes". Blue line- Gradient-based "snakes". Red line- PC-based "snakes". Top row left to right: segmentation at iterations 3, 18, 24. Bottom row left to right: segmentation at iterations 30, 40, 49.

Local Phase Quantization error (LPQe)

From the description of "Local Energy" and "Phase Congruency" [12, 17, 41] it is obvious that various signal features are constructed by the in-phase local Fourier components. Basing on this fact, we have proposed another phase-based edge detection method described in detail in [15]. We have chosen to look at the edge detection problem from a different perspective. We first calculate the local spatial-frequency transform. Then we reconstruct the image, using the unchanged magnitude and the quantized phase. The effect of phase quantization on the reconstruction error is negligible in smooth areas, while it is very significant around edges. The reconstruction error provides therefore an excellent map of the edges and a skeleton of the image in the sense of primal sketch. Figure 12 presents a diagram describing the principles of our method.

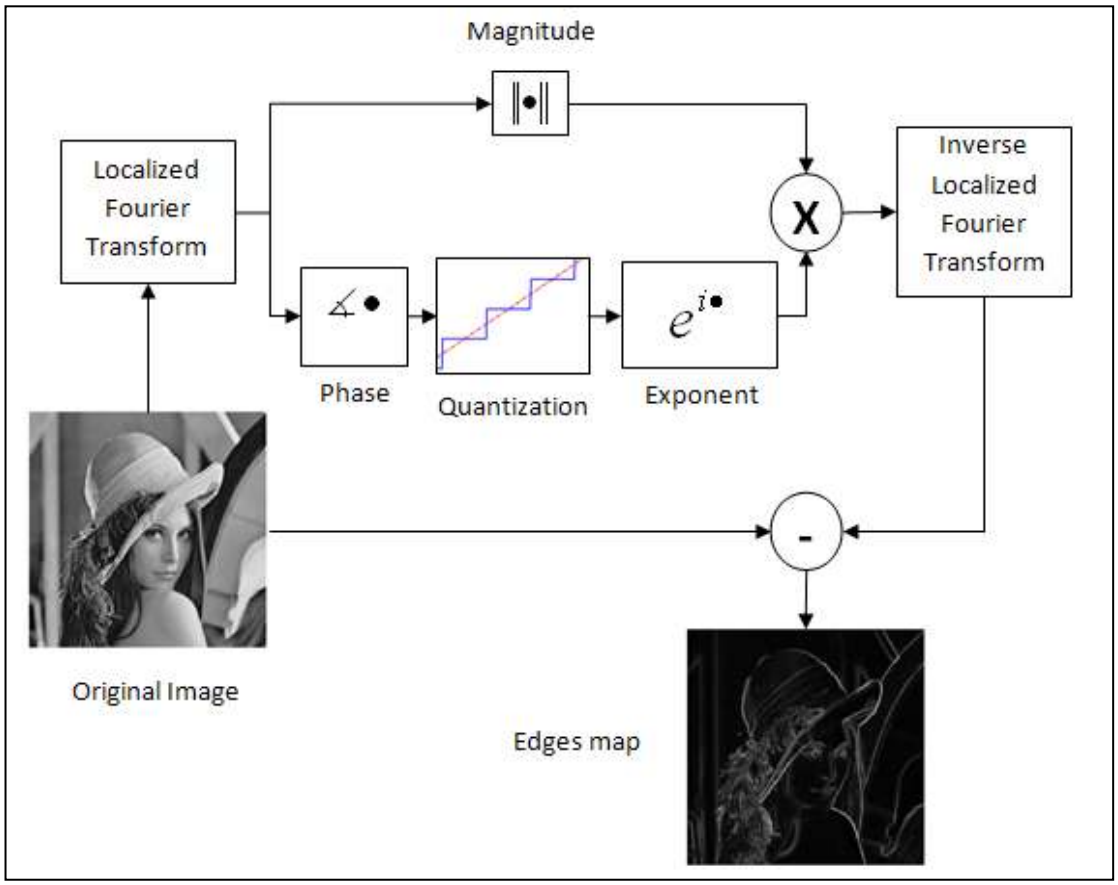


Figure 12: Schematic diagram of LPQe.

This scheme has several possible implementations. The main questions to be asked at this point are: how localized Fourier transform is calculated, and what quantization level should be used. Let us answer each of these questions.

We wish to achieve a maximal error during edges reconstruction, and thus, we will use lowest quantization level- $K_q=2$ (one-bit quantization). The localized Fourier Transform can be calculated by several methods. One of the easiest methods is STFT, in which a window of a preselected form (rectangular, Gaussian, etc…), dimension and size is applied to the signal, with a predefined step. A Fourier Transform is applied to the windowed sub-signal, resulting in a localized Fourier Transform. We have also used a Gabor Filter bank (which is also common spatial-frequency analysis tool) for this goal, but we found STFT to be both easier to understand and implement, while delivering quite impressive results. Let us begin with 1D signals. We will use a rectangular window- of minimal dimensions, 3, and minimal step size, 1, to achieve maximal localization. In this case for each element of the analyzed signal, we will calculate the FT of the element and its two nearest neighbors. After quantizing the resulting phase we will calculate the inverse Fourier transform. The reconstruction error achieved using this method will serve as an edge detector.

Our scheme has the following advantages over gradient based methods: a better ability to deal with noise, and the fact that it does not detect edges in harmonic signals. In the next paragraphs, we will replace the gradient operator by our edge detection scheme, in several applications.

The calculation and representation of STFT for two-dimensional signals demand large memory blocks, as they involve the use of four dimensional matrixes. Thus, in order to keep our scheme simple, in a manner

similar to the 1D case, we will use a 3X3 rectangular window, and a step size of 1 to achieve maximal accuracy in edge detection. The resulting edge maps can be seen and compared with other methods described above in figure 13.

Use following link for additional applications of LPQe derived edges maps in man-mades detection and anisotropic diffusion.

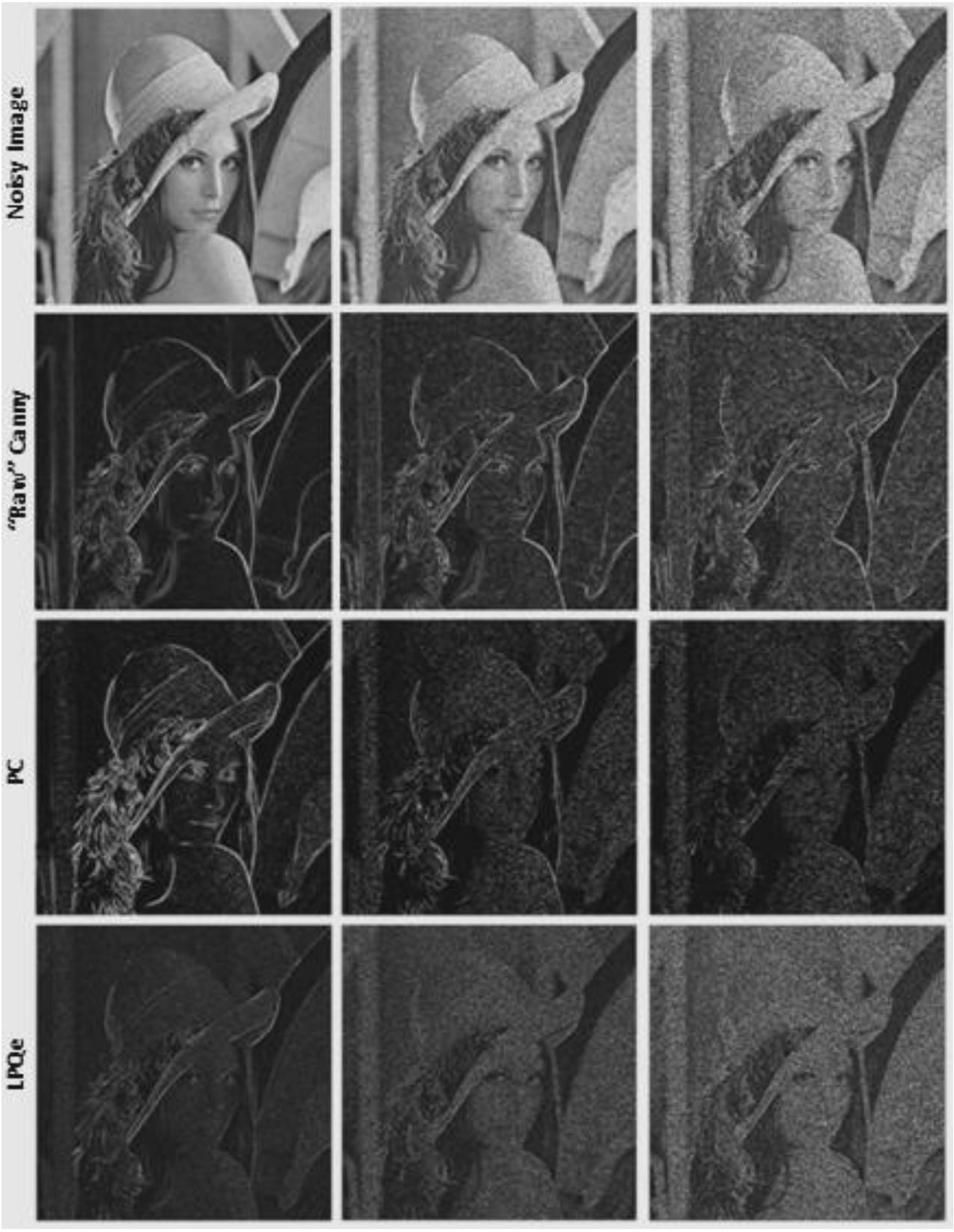


Figure 13: Edges detection in noisy images. Top row (left to right)- noisy images SNR: 20[dB], 10[dB] and 5[dB]. The columns are arranged as follows (top-down): noisy image, Canny, PC and LPQe.

Image primitives via Rotated Local Phase Quantization

Reconstruction of an image from quantized localized phase results in an image with degraded edges. The reconstruction error was therefore utilized by us in the previous section for edges detection. There our goal was to preserve the signal, in edge-free areas, and to degrade it in areas including edges. This is the reason we have chosen the quantization scheme described in [15]. There are several ways to perform angle quantization. The quantization method used by us so far can be described as following:

$$\tilde{\phi} = K\left[\!\left[\frac{\phi}{Q}\right]\!\right], \quad Q = \frac{2\pi}{K} \tag{4}$$

Where the $\bullet$ operator represents the "nearest integer" function, ϕ is the angle of the phase subject to quantization and $\tilde{\phi}$ is the quantized angle.

Thus, for $K_q=2$ (one-bit quantization level) angles between $-\pi/2$ and $\pi/2$ are assigned the value of 0 whereas angles between $\pi/2$ and $3\pi/2$ assume the value of π. This is depicted graphically in the left plane of figure 14, but this is by far not the only possible quantization scheme. An alternative scheme of phase quantization

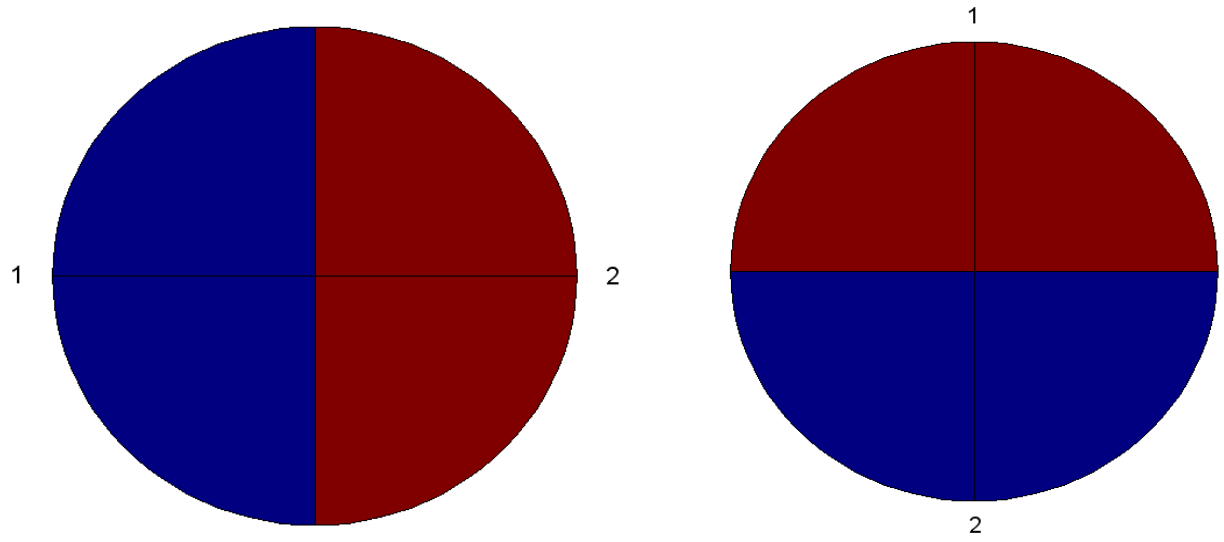


Figure 14: Standart and Rotated phase quantization, $K=2$. Left- quantization to two levels- eq. (4). Right- quantization to two levels rotated by $\pi/2$- eq. (5).

is shown on the right plane of figure 14 and is defined as follows for K_q quantization levels:

$$\tilde{\phi} = K\left[\!\left[\frac{\phi}{Q} - \frac{Q}{2}\right]\!\right] + \frac{Q}{2}; \quad Q = \frac{2\pi}{K} \tag{5}$$

We refer quantization according to eq. (5) as "Rotated Quantization", since it is in a way a rotated version of the "Standard Quantization" used so far.

What are the effects of the alternative phase quantization schemes on the LPQ process? In the edge detection method we have introduced in [15], our goal was to impair the signal, in areas with edges, and to save it intact elsewhere. Thus, when the original signal was real-that is, non-complex, (which is usually the case) we had to guarantee, that the signal resulting from the LPQ process, will be real as well. It is a well known property of the Fourier transform, that if *x(t)* is real the following equation holds:

$$X(w) = X^{*}(-w) \tag{6}$$

This can be equivalently rewritten in a different form, where the conditions on the phase and the magnitude of $X(w)$ are accounted for:

$$\begin{aligned} \left\| X(w) \right\| &= \left\| X(-w) \right\| \\ \measuredangle X(w) &= -\measuredangle X(-w) \end{aligned} \tag{7}$$

We would like to find what are the conditions of the quantization method $Q_{[\,]}(\bullet)$ to result in real reconstructed signal $\tilde{x}_n$ given a real input signal x_n . This is formulated as follows:

$$\begin{aligned} &\left\{ x_n \right\} \in \mathbb{R} \ \forall n \Rightarrow \left\{ \tilde{x}_n \right\} \in \mathbb{R} \ \forall n \\ &X_F = FT\left[x_n \right] \\ &\phi_F = \measuredangle X_F, \ \tilde{\phi}_F = Q \quad (\phi_F) \\ &\tilde{x}_n = IFT \left\| X_F \right\| \bullet e^{i\tilde{\phi}_F} \end{aligned} \tag{8}$$

$$\left\{ x_n \right\} \in \mathbb{R} \ \forall n \Rightarrow X_F = X_{-F}{}^{*} \Rightarrow \begin{cases} \left\| X_F \right\| = \left\| X_{-F} \right\| \\ \phi_F = -\phi_{-F} \end{cases}$$

We find the condition on the Quantization method that will provide real reconstructed signal:

$$\begin{aligned} &\left\{ \tilde{x}_n \right\} \in \mathbb{R} \ \forall n \Rightarrow \tilde{X}_K = \tilde{X}_{-K}{}^{*} \Rightarrow \begin{cases} \left\| \tilde{X}_K \right\| = \left\| \tilde{X}_{-K} \right\| \\ \tilde{\phi}_k = -\tilde{\phi}_{-k} \end{cases} \\ &Q \quad (\phi_k) = \tilde{\phi}_k = -\tilde{\phi}_{-k} = -Q \quad (\phi_{-k}) \Rightarrow \\ &Q \quad (\phi_k) = -Q \quad (\phi_{-k}) \end{aligned} \tag{9}$$

With this reference to eq. (9), we can state the following condition for the LPQ method mapping of a real signal input onto a real signal output. The applied phase quantization must be anti-symmetric:

$$Q \quad (\phi_k) = -Q \quad (\phi_{-k}) \tag{10}$$

The only quantization scheme $Q_{[\,]}(\bullet)$ satisfying this condition is the Standard Quantization model (utilized in [15]), while all other methods will result in a complex signal. This is the reason we had to use the Standard Quantization model in our edge detection scheme. In what follows, we consider the effects of using the Rotated Local Phase Quantization (RLPQ), i.e. LPQ with Rotated Quantization. RLPQ does not satisfy eq. (10) and, thus, results in a complex signal. As such it is composed of a Real and Imaginary components. At this point we have two parameters influencing the RLPQ process:

- Window size- that determines the locality of the quantized phase.
- Number of quantization levels $2 \le K_q < \infty$ (to be precise, the upper bound is finite, and platform-dependent).

Increasing the window size will result in a further degradation of the original signal, as more features located in each window will be affected.

Increasing K_q will reduce the feature (edge) degradation on one hand, and reduce the energy of the imaginary part of the RLPQ signal on the other.

We expect the real component to be similar to the original signal while its features related to phase (such as edges) will be degraded. The magnitude of the imaginary component is related to the rotational nature of the applied quantization, and therefore is influenced by those features.

Blurring via Rotated Local Phase Quantization

Blur is mostly an undesired effect, and is usually caused by motion or "out of focus". The visual effect is of "smearing" and non-sharp appearance of the signal. In the frequency domain, it results in signal attenuation at high frequencies. There are several ways to introduce blur effects:

- Box blur- introduced by convolution with a rectangular window, or Low Pass (LP) filtering.
- Gauss blur- introduced by convolution with a Gaussian window, or by multiplication with a Gaussian in frequency domain.
- Motion blur- can be introduced by multiplication with a matrix obtained by circular shift of a vector starting with several ones, and trailing zeros, or via convolution with appropriate PSF (Point Spread Function).

Although blur is mostly an undesirable effect and many algorithms have been proposed for deblurring. Yet, blur can be handy in some cases: artistic effects, reduction of image noise or detail, image enhancement in scale-space analysis, etc...

As mentioned above, blur is usually achieved through convolution with an appropriate PSF or LP filtering, where the high frequencies of the signal are attenuated. We wish to achieve blurring through a controlled localized phase degradation, namely quantization.

Let us begin our analysis with a synthetic 1D signal. We compare the signal blurred by RLPQ with that blurred by box-filter. In addition, we present the spectrum of each signal. The signal subject to blurring is

composed of several types of edges: a roof, a step and a delta. We apply an increasingly growing window width, to introduce a stronger blur effect, both with Box and RLPQ. As the imaginary part of RLPQ signal has a lower level of energy compared to the original signal, we linearly rescale it, to get comparable results. The *Kq* value used in the following figures is 2, but the effect introduced is common to other *Kq* values as well. Figure 16 presents blurring achieved via RLPQ on an image.

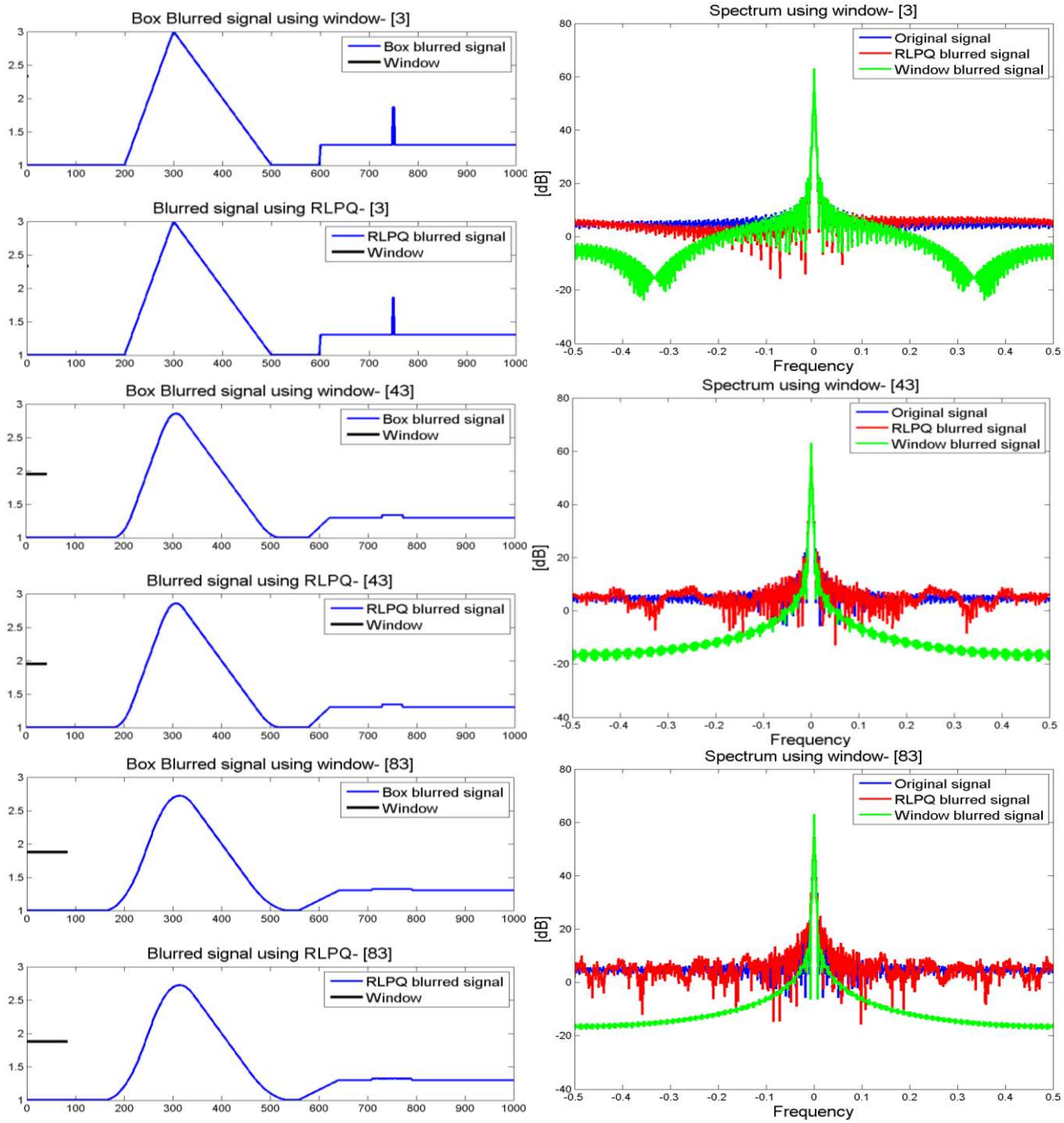


Figure 15: Blurring methods- RLPQ vs. Box filtering applied to 1D signals. Left column- original signal, blurred by Box Blur (top) and RLPQ (bottom) for an increasing window size; in top-to-bottom order-[3, 43, 83]. Right column- the spectra of original signal-blue, "Box blurred" signal-green, and RLPQ blurred signal (not linearly stretched)- red. The RLPQ signal was linearly amplified, for higher similarity.

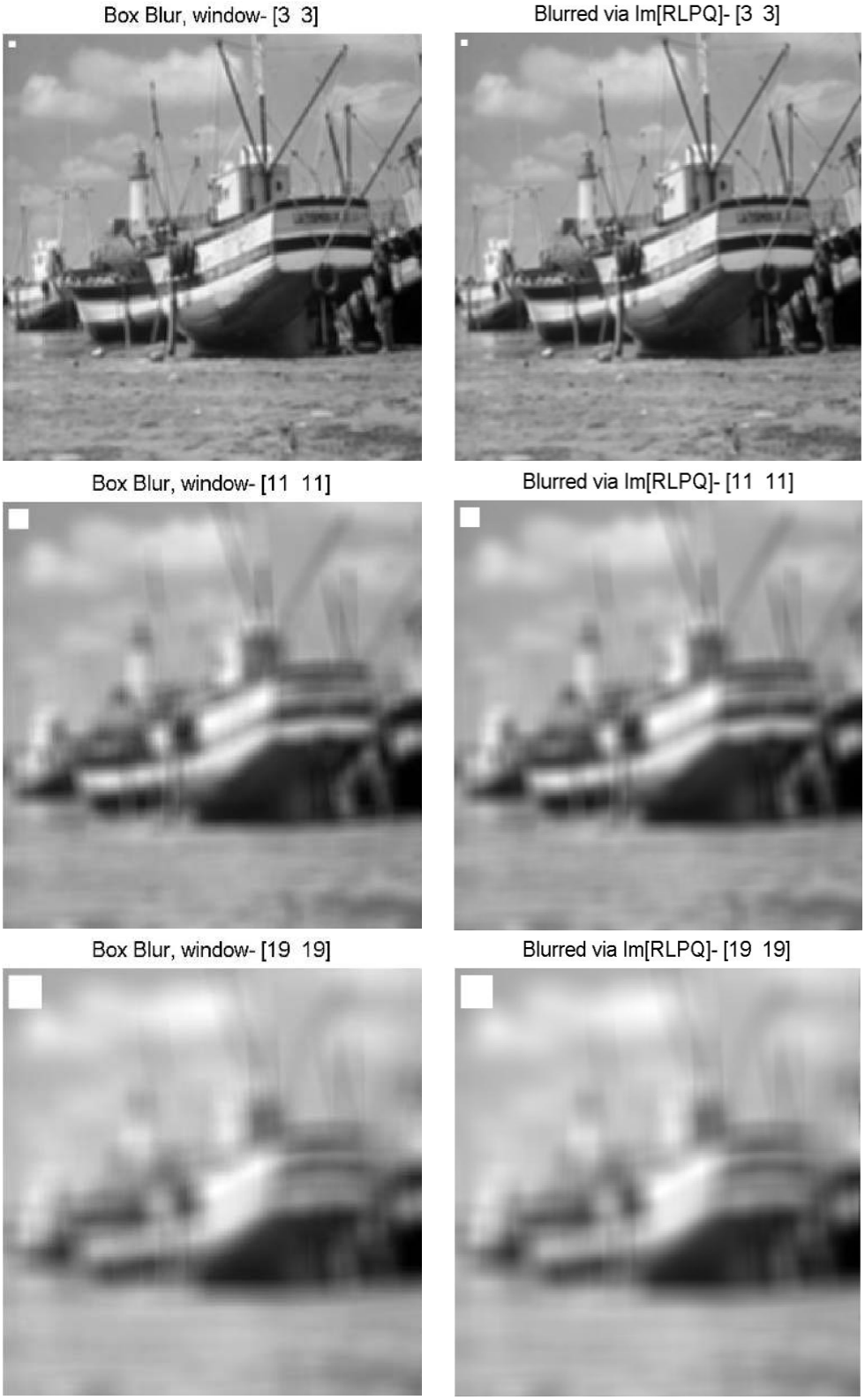


Figure 16: Blurring methods- RLPQ vs. Box filtering applied to 2D signals. Left column- "Boat" image, processed by Box Blur with an increasing window size; in top-to-bottom order- [3X3, 11X11, 19X19]. Right column- "Boat" image, blurred by RLPQ with same window sizes. White squares in top left part of images present the dimensions of the applied window.

We have mentioned at the beginning of the chapter, that the RLPQ method has two parameters- window size and quantization level-K_q. We have demonstrated the influence of the window sizes in **Error! Reference source not found.** and figure 16. K_q controls the energy ratio between the Real and Imaginary parts if RLPQ.

Visually, the blurred signal achieved is similar to the box blurring results. The method described above, indeed achieves signal blurring through RLPQ. As far as we know, the above scheme is novel, and is presented here for the first time. Though our method is more demanding computationally, it presents an original way to achieve the blurring effect.

Edge detection via Rotated Local Phase Quantization

As we have already mentioned before, edges are very important image primitives. The cartoon for example is merely an edges map, with some constant intensity fill between edges. The next step is to replace the constant intensity area with a texture. Assuming that these textures are at a sufficient amount of variability and complication level, an image can be constructed using the above image primitives as building blocks. We wish to demonstrate here, that the Real component of the RLPQ signal is an image primitive – an edges map for $K_q=2$, a cartoon for $K_q=3$ and more complicated images for higher quantization values. As the K_q value will increase, we expect the Imaginary part of the RLPQ to drop, in favor of the Real part of RLPQ. In addition we expect that the Real part of RLPQ will become more and more similar to the original image. The cause is that higher quantization level reduces the degradation of the reconstructed signal.

The following figures demonstrate several properties of the Real component of RLPQ :

- For $K_q=2$ it detects edges, at various window sizes. Higher window size introduces some artifacts, thus a good edge detection ability is demonstrated for larger windows, as opposed to LPQe .
- K_q values higher than 2 result in a cartooning effect, which decreases as the quantization value grows. Larger window creates aura around edges in the cartoon image.
- Degradation around edges, increasing with window size can be seen in figure 17. It visually resembles "ringing effect" and can be explained in the following manner: larger windows allow penetration of image features such as edges to pixels on a longer distance from the features. Those features are primary affected by phase impairment, which results in reconstruction error. This error produces the visual effect. This explains why small windows to be used for quality results.

Both detected edges and cartoonized images are building blocks/primitives of images, and in addition are part of various image segmentation schemes. Therefore the demonstrated above properties of RLPQ are of particular interest.

Figure 19 describes graphically, how image primitives are created from the Real part of RLPQ.

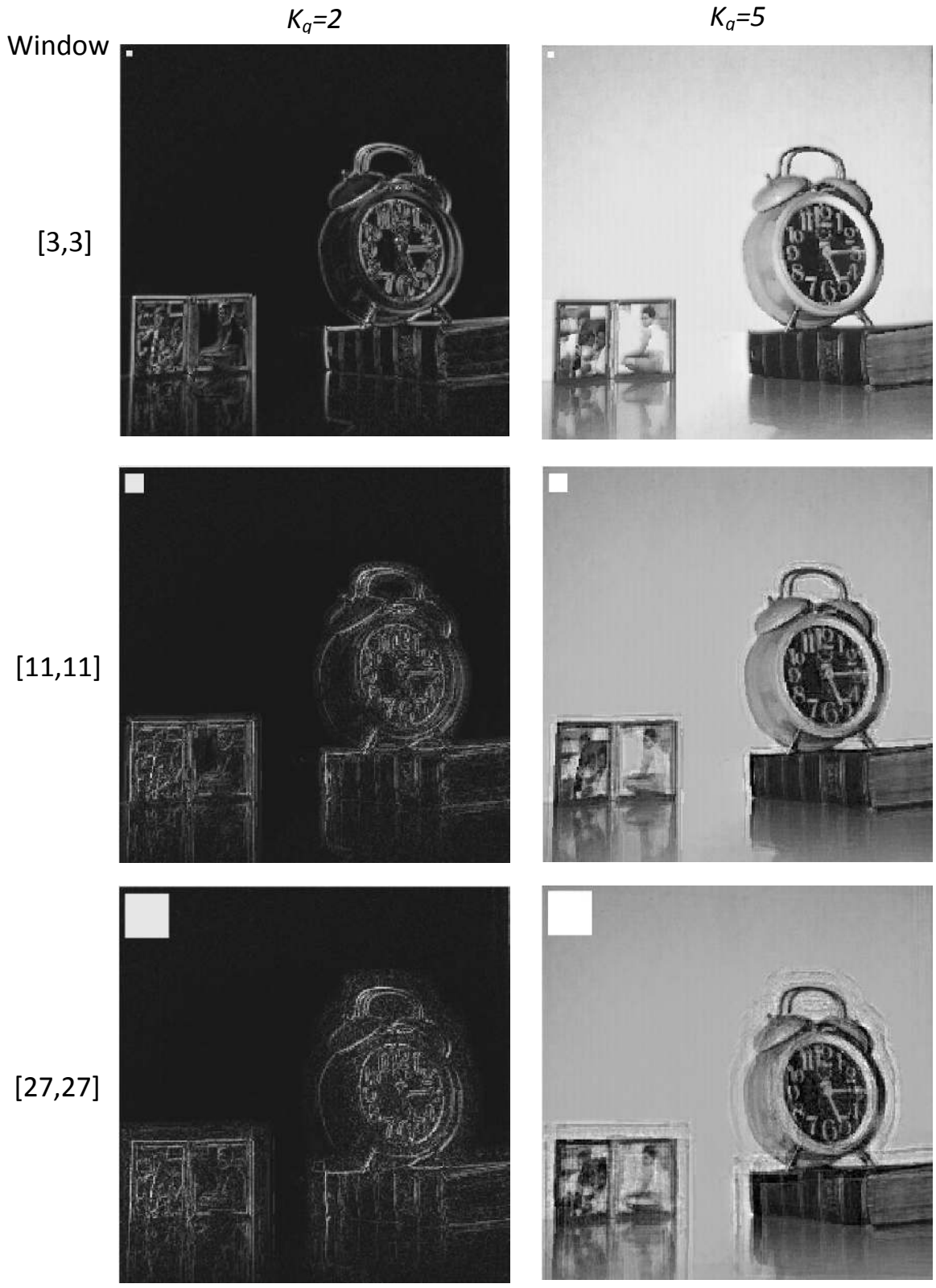


Figure 17: K_q and window size impact on Re[RLPQ]. Left column- Real RLPQ (K_q=2) component of "Clock" image, with an increasing window size; in top-to-bottom order-[3X3, 11X11, 27X27]. Right column- Real RLPQ (K_q=5) component of "Clock" image with same window sizes. Note that larger window results in image degradation, nevertheless, edge detection can be seen for relatively big windows (K_q=2), as opposed to LPQe method.



Figure 18: K_q impact on image primitives detection via Re{RLPQ}. Row wise- Lena Image primitives detection from Real RLPQ for increasing K_q values: [2, 3, 5, 16].

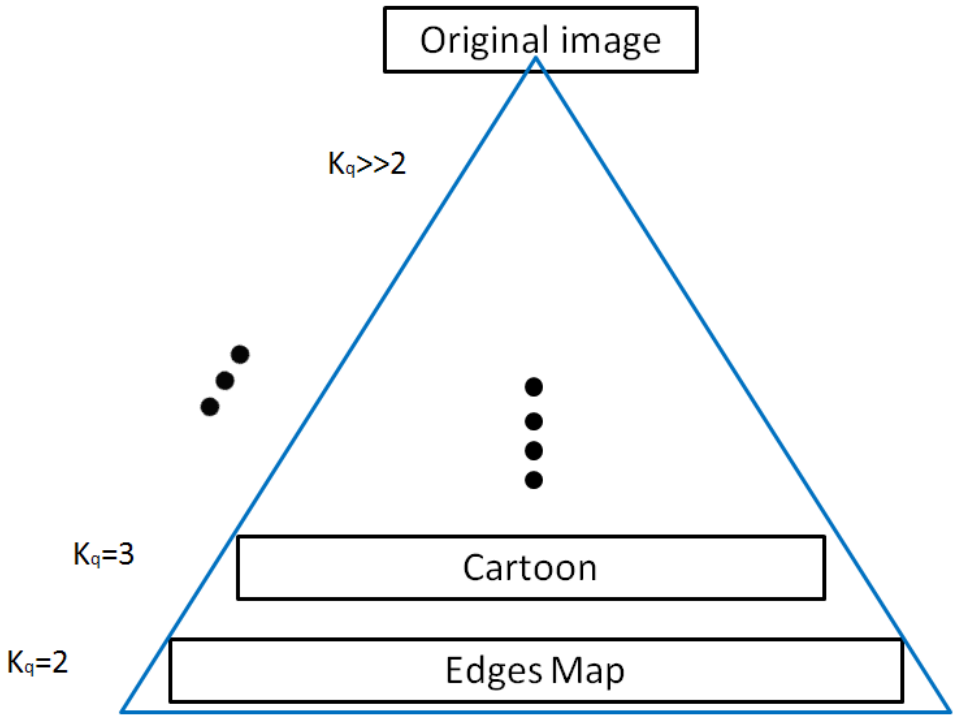


Figure 19: Image primitives piramid from Re[RLPQ]. Low K_q values present image primitives such as edges and cartoons, higher K_q value result in more and more complicated image, until it is indistinguishable from the original.

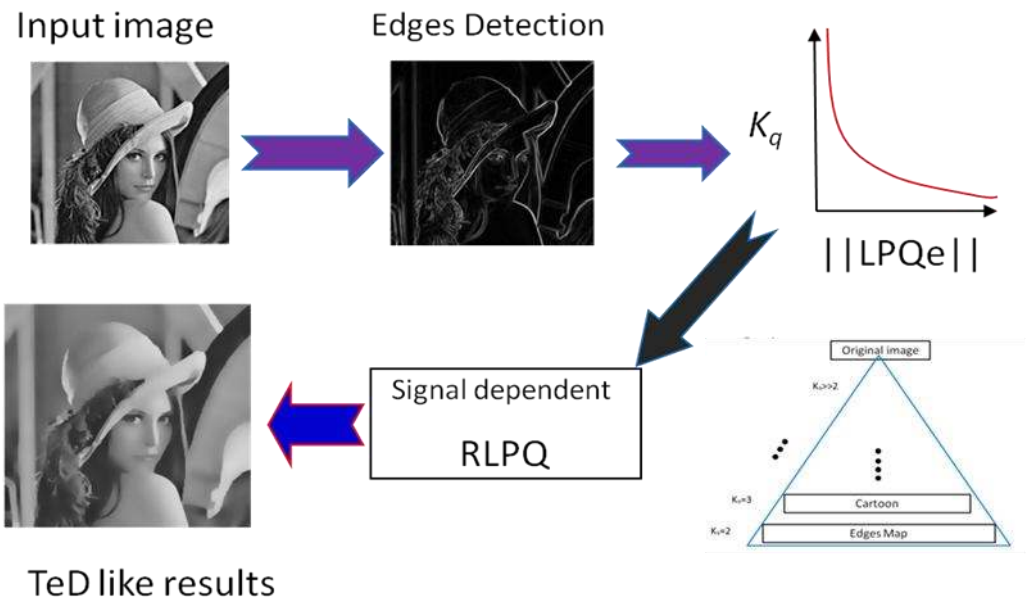


Figure 20: Edge preserving and signal dependent *Kq* definition. *Kq* value is derived from signal edges map. Also note *Kq* is not restricted to integer values.

The iterative diffusion schemes described in [5, 6, 45] have been proven to be a useful tool in image processing. Presenting an image by means of an appropriate physical model grants the researchers a rich mathematical background together with intuitive insights regarding the applied process. Diffusion equations such as the heat and Telegraph diffusion (TeD) were proposed for various tasks such as image segmentation, noise suppression, edge sharpening and super resolution. The hypothesis that similar results can be achieved through localized phase manipulations seems to be very promising, and is our prime research lead for the nearest future.
Up until this point we have used constant *Kq* value for the whole signal. We have further improved the way quantization is implemented in RLPQ scheme. Now *Kq* value is signal dependent, and edge preserving (higher values around edges). It is also worth mentioning that *Kq* value is not restricted to integers, which allows further flexibility of our method.



Figure 21: RLPQ and TeD results similarity. Left - Lena image after application of RLPQ-based iterative scheme with signal depended *Kq* value. Right- Lena image after Telegraph Diffusion method application.. Right column images received from authors of [5, 6].

As can be seen from the above images, the current results achieved from RLPQ iterative schemes and TeD and FaB TeD, both similar to a result of a diffusion process. However the current results are far from being identical. We must note that while the diffusion schemes are iterative by nature, our scheme result is archived in "one shot"- single application of the RLPQ- which is naturally less demanding in computation effort and time invested. We believe that identical results for TeD and RLPQ can be archived, but it is not an easy task due to the great difference between the schemes. We must mention that the result demonstrated on the left plane of figure 21 is archived by utilization of signal dependent *Kq* (see figure 20 for schematic explanation).

Discussion and Future Research

Phase is a non-intuitive feature, an attribute that causes it to be overlooked too often. In this research we have shown that phase is extremely significant for a high quality feature extraction and analysis for both one-, and two-dimensional signals. We have demonstrated that localized phase manipulations results in unpredicted and useful effects, that were achieved till now by other means such as Low/High Pass filtering or iterative schemes.

During our research, we have been using the very basic tools such as rectangular windows and time frequency analysis schemes, in order to show the robustness of our approach.

The methods we have proposed in this report are not only original, but in some cases worthwhile in terms of the reduction of the computational effort. We believe that the ideas presented here, are therefore both of theoretical as well as practical value and interest, and that further research should be carried out.

2D PC via 2D AS

The phase congruency method for edge detection has combined both important insights and good performance. Unfortunately, as no single 2D Hilbert Transform definition exists, the 2D PC scheme proposed in [12] is achieved through multiple projections to 1D. We have tried to find an alternative truly 2D definition of 2D PC, using the 2D HT definition proposed in [42, 43]. This task has not been completed however, it is very important as it has the potential to be an excellent edge detector.

Edge detection by GEF elements impairment

We have assumed that the impairment of Gabor elements a_{mn} will result in a reconstruction error. A controlled a_{mn} elements impairment was supposed to give us sufficient tools for signal analysis, in a manner similar to what we have described in [15] . Unfortunately, due to unexpected implementation issues (see details using the following link), we have failed in applying the Generalized Gabor Scheme to our research. We do believe that proper implementation which must involve combination of more efficient algorithms (like FFT in FT) and computers with superior memory and processing abilities will propose a fruitful field for research.

Quantization of Wavelets phase

The Wavelets analysis is a widely used localized scheme. While we have used Fourier based analysis schemes, we can think of extending our approach to various mother Wavelet functions, as their natural readiness for scale-space analysis may be beneficial. According to [46] the Wavelet phase carries much information, which allows better reconstruction than the one achieved from Fourier phase. We believe that Wavelet phase manipulations, in a manner similar to the one we have described in [15] may provide useful results. The fact that there are various implementation schemes of signal analysis by means of Wavelet Transform permits their easy implementation and efficient schemes. We are currently performing a research using dual-tree wavelets [47].

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